

What Changes, and Where?

Legislative Churn Across Saudi Arabian Legislation

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Abstract

Fifty-eight consecutive articles of the Saudi Law of Sharia Procedure were repealed by a single royal decree in December 2021, and seventy-five of its 243 articles — nearly a third — are now repealed in total. The instrument is still in force, still cited, and unchanged in size and numbering. How common is that? Legal systems are usually studied as they stand at a moment, because the evidence to study them as they move has not existed. Using a machine-readable corpus in which 13,089 articles across 272 Saudi legislative instruments carry a verified legal status, this article measures what changes. Some 973 articles — 7.4 per cent — are no longer in their original form: 730 amended, 138 repealed, 105 added. That change is very unevenly spread. Nearly three-fifths of instruments record no change at all, and the Gini coefficient of changed articles across instruments is 0.82. Instruments change in characteristically different ways, so that a single count of ‘amendments’ conceals more than it reveals: one instrument is hollowed out by repeal, another accretes new articles, a third is rewritten in place. Two findings connect this to how a legal system is used. Instruments that other instruments cite change substantially more than instruments nobody cites — 10.7 per cent against 6.1 per cent, rising to 13.4 per cent for the fifteen most cited — although age is not controlled for and may explain some of it. And cross-instrument references are more than twice as likely as the base rate to point at an article that has since changed, while an instrument’s references to its own articles are not: drafters maintain their own internal references and cannot maintain anyone else’s.

Keywords: legislative amendment; consolidation; statutory maintenance; Saudi Arabia; legal corpora; empirical legal studies

1 Introduction

On 26/5/1443H — 30 December 2021 — a single royal decree issued the Saudi Evidence Law, 129 articles in eleven chapters.¹ The same decree repealed articles 101 to 158 of the Law of Sharia Procedure: fifty-eight consecutive articles, the whole of that instrument’s treatment of proof, lifted out and re-enacted elsewhere. Two later decrees repealed ten more. Seventy-five of the Law of Sharia Procedure’s 243 articles are now repealed — nearly a third of the instrument — and none of this is visible in its shape. It remains in force, its articles are still numbered 1 to 243, and a reader who does not check the status of each article will not discover which of them still say anything.

¹Evidence Law (Royal Decree M/43, 26/5/1443H).

This is not a defect peculiar to one statute. It is what a working legal system looks like from the inside. Statutes are amended, articles are repealed, new articles are inserted, and the instrument a reader holds is a composite of enactments made across decades. Every legal system does this. What no legal system has been able to say, until recently and for very few jurisdictions, is how much of it does it, where, and how far.

The obstacle is evidential. Measuring change requires knowing, for each article, whether it is the article originally enacted — and legal databases usually present the consolidated current text, silently. The history is either absent, or present as prose a human must read. We are aware of no Arab jurisdiction for which the question has been asked at corpus scale, and of few studies anywhere that ask it at the level of the individual article.

This article asks it for Saudi Arabia. It uses a corpus in which every article verified against an official source carries a legal status — original, amended, repealed or added — covering 13,089 articles across 272 instruments.² Saudi Arabia is a good case for the same reason it was a good case for studying the definitional lexicon: its legislation was substantially rewritten during the 2020s, so instruments enacted decades apart are being maintained side by side.

Four findings follow.

First, the aggregate is small and the distribution is not. Some 973 articles, 7.4 per cent of the corpus, are no longer in their original form. But 58.8 per cent of instruments record no change at all, and the Gini coefficient of changed articles across instruments is 0.82. Saudi legislation does not age evenly; a small number of instruments absorb almost all of the movement.

Second, instruments change in characteristically different ways. The Sharia Procedure Law is dominated by repeal, the Implementing Regulation of the Commercial Agencies Law almost entirely by addition, the VAT Regulation by amendment in place. A single figure for ‘amendments’ would put these three in the same bucket, and they are not the same phenomenon: one is a jurisdiction being transferred out, one is a regime being extended, one is a regime being tuned.

Third, change concentrates where the system leans hardest. Instruments that other instruments cite have a mean churn rate of 10.7 per cent against 6.1 per cent for instruments nobody cites, rising to 13.4 per cent for the fifteen most cited. Section 7 sets out why this is suggestive rather than established: age is not controlled for, and it cannot be, because the corpus dates only two of its instruments.

Fourth — and this is the finding with the clearest practical edge — cross-instrument references are disproportionately exposed to change. Seventeen per cent of the references from one instrument to a specific article of another point at an article that has since been amended or repealed, against a corpus base rate of 7.4 per cent. References an instrument makes to its own articles show no such excess. The asymmetry has an obvious mechanism: a drafter amending an instrument can see and fix the references inside it, and has no way to find the references to it.

Section 2 describes how amendment works in Saudi practice and why the article is the right unit. Section 3 sets out the data and the two measurement traps in it. Sections 4 to 7 report what was found, and sections 8 and 9 draw out what follows and what the method cannot see.

²The corpus and the scripts that reproduce every number and figure here are openly archived; see the data availability statement.

2 Amendment in Saudi legislative practice

2.1 How an instrument changes

A Saudi law is enacted by royal decree approving a Council of Ministers resolution; an implementing regulation is issued by the competent minister or authority under a power the law confers. Change arrives the same way. A later decree or resolution amends named articles of the earlier instrument, substituting wording, adding an article, or repealing one. The amending instrument is generally not itself a standing part of the legal furniture: it does its work and the consolidated instrument absorbs it.

Three consequences follow for measurement.

The unit of change is the **article**, not the instrument. An amending decree names articles, and the resulting record is article-level. Counting amended *instruments* would compress an instrument in which one article was retouched and an instrument in which forty were rewritten into the same observation.

Repeal is often **partial and in place**. When an article is repealed without the instrument being replaced, the article number is retained and the provision emptied, so that the numbering of everything after it survives — which is why the Sharia Procedure Law can lose seventy-five articles without being renumbered. The instrument’s apparent size does not fall.

And the record is **consolidated by default**. The official portal shows the current text; the history of an article, where it is shown at all, sits behind it. A reader who does not look for the history will not encounter it, which is precisely why the aggregate question has not been answerable.

2.2 Why this has not been measured

The empirical study of legal systems has developed measures of size, of complexity and of citation structure.³ Work that follows a legal system over time has generally done so at the level of the instrument or the code section count, because that is what the available data supports. Machine-readable legal corpora have grown considerably, but they are assembled to train language models and hold consolidated running text rather than the status of its parts.⁴ Structural standards can represent amendment faithfully, but representing it is not the same as having the data.⁵

The drafting literature, for its part, treats amendment as a craft question — textual amendment against non-textual, the case for consolidation, how to avoid leaving the statute book unreadable.⁶ Its recommendations are sound and its evidence is illustrative. The gap this article addresses is between the craft advice and any measurement of what the craft actually produces at system scale.

³Daniel Martin Katz and Michael J Bommarito, ‘Measuring the Complexity of the Law: The United States Code’ (2014) 22 *Artificial Intelligence and Law* 337; Corinna Coupette and others, ‘Measuring Law Over Time: A Network Analytical Framework with an Application to Statutes and Regulations in the United States and Germany’ (2021) 9 *Frontiers in Physics* 658463.

⁴Peter Henderson and others, ‘Pile of Law: Learning Responsible Data Filtering from the Law and a 256GB Open-Source Legal Dataset’ (NeurIPS 2022, Datasets and Benchmarks Track); Joel Niklaus and others, ‘MultiLegalPile: A 689GB Multilingual Legal Corpus’ (Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, 2024).

⁵Monica Palmirani and Fabio Vitali, ‘Akoma-Ntoso for Legal Documents’ in *Legislative XML for the Semantic Web* (Springer 2011) 75.

⁶Helen Xanthaki, *Thornton’s Legislative Drafting* (6th edn, Bloomsbury Professional 2022) ch 12; Reed Dickerson, *The Fundamentals of Legal Drafting* (2nd edn, Little, Brown 1986).

3 Data and method

3.1 The status layer

The corpus records, for each article it verified against an official source, a legal status in the terms the source uses: *asliyyah* (original), *mu'addalah* (amended), *mulghah* (repealed), *mudafah* (added). The layer covers 13,089 articles across 272 of the corpus's 291 tracks. For 972 articles it also records an amendment history.

Every measure below rests on that status field, so its provenance matters: it is transcribed from the official record for the article, not inferred from the text. Where the corpus could not establish a status, the article is counted in neither the numerator nor the denominator.

3.2 Two shapes of amendment history, and why conflating them fails

The amendment history comes in two shapes, and the difference is not cosmetic.

In the **event-log** shape, each entry names an amending decree and carries no text. The number of entries is then the number of amendment events the article has been through: 617 articles are recorded this way, of which 109 have been amended more than once and one has been amended seven times.

In the **prior-version** shape, an entry carries the superseded wording, so the current text can be compared against what it replaced.

The trap is that not every entry carrying text is a prior version. Measured against the article's current wording, entries labelled 'amended' or carrying no label at all have a median Jaccard similarity of 1.00 — they restate the text as it now stands. Only entries labelled 'original' are genuinely superseded wording, with a median similarity of 0.82. Treating every text-bearing entry as a prior version would have produced a magnitude analysis over 226 articles reporting that amendments barely change anything, which is an artefact of the schema rather than a fact about Saudi law. The analysis script publishes the similarity evidence for the exclusion rather than asserting it, and the magnitude measure in section 6 is confined to the 87 articles that carry a genuine prior text.

3.3 Measures

Churn is the share of an instrument's articles that are not original — amended, repealed or added. Concentration across instruments is reported as the share of instruments with no recorded change, the share of changed articles held by the ten most changed instruments, and the Gini coefficient of changed articles.

Magnitude is the Jaccard similarity between an amended article's current wording and its superseded wording, over normalised Arabic content tokens, using the same normalisation as the companion study of the definitional lexicon — diacritics and *tatweel* stripped, the *hamza* forms of *alif* folded to bare *alif*, *teh marbuta* to *heh*, *alif maqsura* to *yeh*.

Exposure is the share of cross-references that resolve to a specific article whose status is known and which is amended or repealed. Every such share is reported against the corpus base rate of changed articles, so that a number close to the base rate cannot be mistaken for a finding.

Measure	Value
Articles with a recorded legal status	13,089
Instruments covered	272
Original	12,109
Amended	730
Repealed	138
Added	105
Articles not in their original form	973 (7.4%)
Instruments recording no change at all	160 (58.8%)
Share of changed articles in the ten most changed instruments	38.4%
Gini of changed articles across instruments	0.82
Distinct amending instruments identified	179
Articles amended more than once (event-log subset)	109

Table 1: Legislative churn in the corpus.

4 The shape of change

Table 1 gives the aggregate. Three features deserve comment.

The total is modest. Of 13,089 articles, 12,109 stand as enacted. That 7.4 per cent of a legal system’s text has moved is neither alarming nor negligible; it is the sort of figure that only becomes interesting when disaggregated.

The distribution is extreme. A clear majority of instruments — 160 of 272 — record no change whatever, and the ten most changed instruments hold better than a third of all changed articles. A Gini coefficient of 0.82 puts the concentration of legislative change on a par with the concentration of citation reported for the same corpus, and the two concentrations turn out to be related (section 7).

Amendment is itself concentrated. 179 distinct decrees and resolutions account for the amendments that could be attributed, and the ten most active account for 35.4 per cent of the article-amendment pairs. One royal decree alone touches 81 articles. Legislative change in this system arrives in occasional large consignments rather than as a steady drip — a shape that matters for anyone maintaining a downstream copy of the law, since it means the maintenance burden is bursty rather than continuous.

5 Instruments change in different ways

Figure 1 separates the twelve most changed instruments into amendment, repeal and addition, and the separation does real work. Three signatures are visible, and they are three different legal events.

Hollowing out. The Law of Sharia Procedure has 90 changed articles, of which 75 are repeals — 54 per cent of every repealed article in the corpus, in one instrument. Fifty-eight of those repeals are the single block of articles 101 to 158 removed by the decree that issued the Evidence Law. The rest came from elsewhere, and the breakdown is a useful illustration of the limits of the record: nine articles were repealed by a later decree, one by an earlier one, and for seven the corpus records the

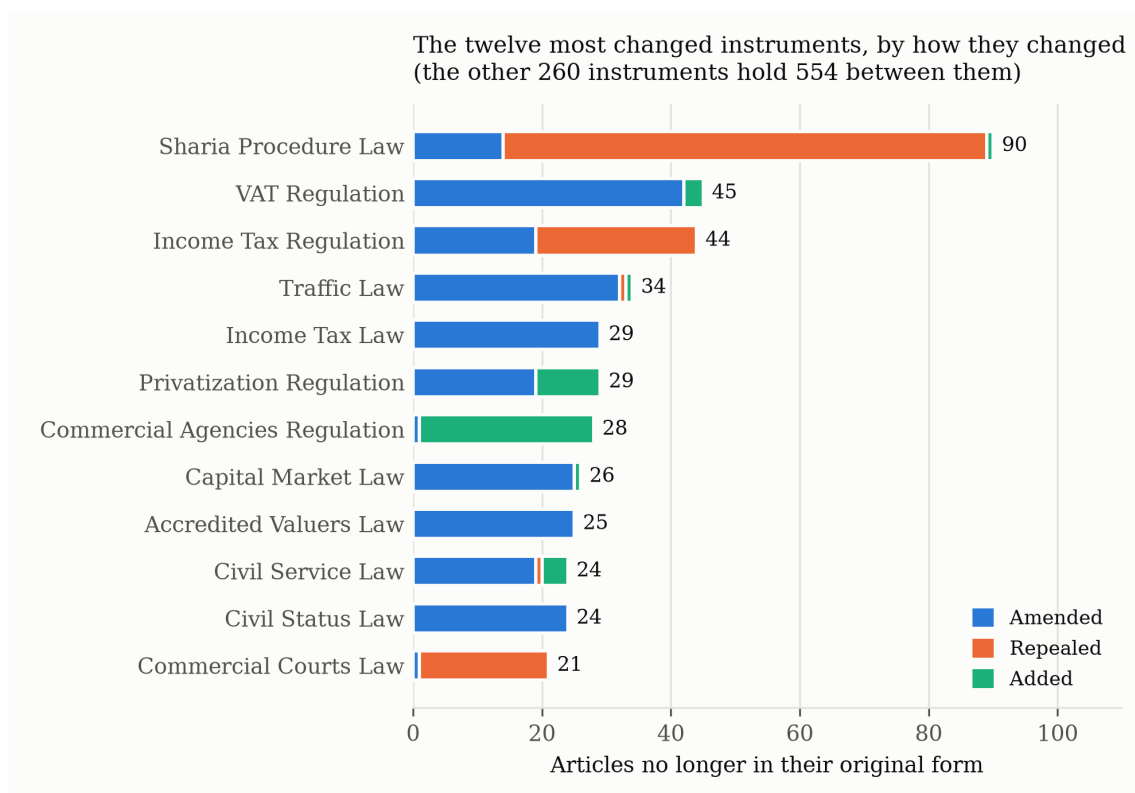


Figure 1: The twelve most changed instruments, split by how they changed. The split is the point: the same total is reached by repeal in one instrument, by addition in another, and by amendment in place in a third.

repeal but no repealing instrument at all.⁷ The Commercial Courts Law shows the same signature at smaller scale: 20 of its 21 changed articles are repeals. This is what it looks like when a subject matter is moved out of an instrument and into a new one, leaving the original in force but emptied in the relevant part. An instrument in this condition is the most dangerous kind for a reader, because its size, its numbering and its continued force all suggest that it is intact.

Accretion. The Implementing Regulation of the Commercial Agencies Law has 28 changed articles, of which 27 are additions and one an amendment. Nothing was taken away; a regime was extended in place. The Implementing Regulation of the Privatization Law shows a milder version, with 10 additions against 19 amendments.

Tuning. The VAT Regulation has 42 amendments, three additions and no repeals; the Income Tax Law, 29 amendments and nothing else; the Civil Status Law, 24 amendments and nothing else. Here the instrument's structure is stable and its content is being adjusted — the pattern one would expect of a technical regime being kept current.

The fiscal instruments deserve a note of their own. Three of the twelve most changed instruments are tax instruments — the VAT Regulation, the Income Tax Regulation and the Income Tax Law — and between them they hold 118 changed articles, an eighth of every changed article in the corpus.

⁷Royal Decree M/43 (26/5/1443H) accounts for 58 of the 75 — articles 101–158; Royal Decree M/101 for 9 — articles 227–235; Royal Decree M/93 for article 35. Seven further articles are recorded as repealed with no decree attributed.

That is consistent with what one would expect of a system that introduced value added tax in 2018 and has been adjusting the surrounding regime since, and it is a reminder that a churn ranking is partly a map of recent policy attention rather than of anything intrinsic to the instruments.

6 What amendment does to the text

For 87 amended articles the corpus records the wording that was replaced. That is 11.9 per cent of all amended articles, and the sample is not random: it is the articles whose official record happened to expose a prior version. Nothing below should be read as an estimate for the other 88 per cent.

Within that sample, the median similarity between the current and the superseded wording is 0.82, and the median change in length is four tokens. Thirty-two of the 87 are near-identical (similarity at or above 0.90), thirty-eight are moderate revisions, and seventeen are substantial rewrites (below 0.50). Forty-eight articles grew and thirty-two shrank.

The distribution matters more than the median. Most amendment in this sample is small — a phrase substituted, a cross-reference updated, a threshold changed. But roughly one amendment in five falls below 0.50, which for articles of similar length means that about a third of the vocabulary has gone.⁸

The extreme cases are extreme in substance and not only in wording. Article 72 of the Law of the Judiciary shrank from 29 tokens to 6 and shares three of them with its predecessor. It formerly provided that the Deputy Minister of Justice be chosen from among serving or former judges of a stated minimum grade; it now provides only that the post is appointed at the excellent rank. The qualification requirement was not reworded — it was deleted, and the similarity score is the trace of that deletion rather than its measure.

An amendment, in other words, is not a category with a typical magnitude. It is a label covering interventions that range from a substituted preposition to the removal of a substantive condition, and a system that tells a reader only that an article ‘has been amended’ has told them very little.

7 Change and reliance

7.1 Instruments others rely on change more

Restricting to the 189 instruments with at least 20 articles — below that a churn rate is dominated by its denominator — the instruments that other instruments cite have a mean churn rate of 10.7 per cent, against 6.1 per cent for the 115 instruments no other instrument cites. The fifteen most cited average 13.4 per cent (Figure 2).

The gradient is clean, and it is also exactly the sort of result that deserves scepticism. **Age is not controlled for.** An instrument enacted in 1426H has had two decades in which to be amended and two decades in which to accumulate citations; an instrument enacted last year has had neither. The gradient is therefore consistent with a real relationship between reliance and change, with simple ageing, or with both. It cannot be separated here: the corpus registry carries a publication date for

⁸Jaccard similarity is the intersection over the union, so for two versions of equal length a score of 0.50 corresponds to roughly two-thirds of tokens in common. A low score is a substantial revision; it is not a synonym for ‘most of the article was replaced’.

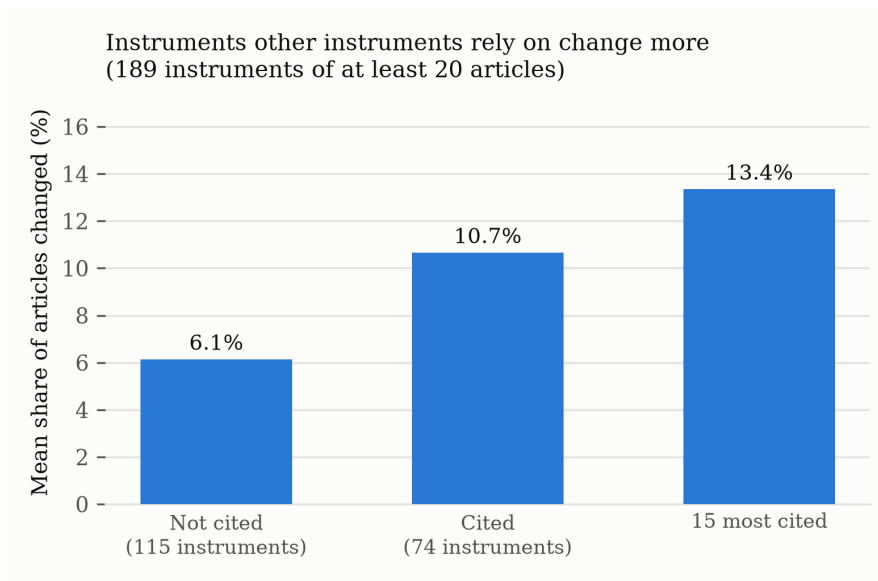


Figure 2: Mean churn rate by citation tier, over the 189 instruments of at least 20 articles. The three tiers are not disjoint: the fifteen most cited are a subset of the cited.

Reference type	Resolved	At changed text	Share
To an article of another instrument	86	15	17.4%
To an article of the same instrument	2,538	218	8.6%
Corpus base rate of changed articles	—	—	7.4%

Table 2: Exposure of cross-references to changed articles.

two of its 291 tracks, so the control is unavailable. Establishing which explanation holds would need a dated corpus, and building one is a worthwhile piece of work in itself.

What can be said without the control is the descriptive fact, and it is not trivial: the instruments a legal system’s other instruments depend on are, as things stand, the least stable part of it. Whatever the causal story, a reader or a system that relies on the Companies Law or the Labour Law is relying on text that moves more often than the average.

7.2 Cross-instrument references are exposed; internal ones are not

Of the corpus’s cross-references, 2,624 resolve to a specific article whose status is known. Of those, 233 — 8.9 per cent — point at an article that has since been amended or repealed. Against a corpus base rate of 7.4 per cent that is a ratio of 1.19: essentially nothing. Reported alone, ‘nine per cent of citations point at changed law’ would be a finding about the base rate wearing a disguise.

The result appears when the references are split by type.

References an instrument makes to its own articles sit at the base rate: 8.6 per cent against 7.4. References it makes to another instrument’s articles run at 17.4 per cent — twice the internal rate and 2.4 times the base rate.

The mechanism is not mysterious, and that is what makes the result useful. A drafter amending an instrument has the whole of that instrument in front of them. Internal cross-references are visible, and updating them is part of the job. References *into* the instrument from elsewhere are not visible

— there is no index of them, and no step in the process at which anyone is asked to look. The excess exposure is the measurable trace of a gap in the drafting workflow rather than of anyone’s carelessness.

Two cautions. The inter-instrument figure rests on 86 resolved references, which is a small base; the corresponding confidence is correspondingly loose, and the finding should be treated as an indication with a plausible mechanism rather than a settled rate. And ‘points at an amended article’ is not the same as ‘is now wrong’: an amendment may leave the cited proposition untouched. What the number identifies is a set of references that nobody has checked, not a set of errors.

8 What follows

Consolidation is not the whole answer. The standard remedy for an unreadable statute book is consolidation — publishing the current text so that a reader need not assemble it. Saudi practice already does this: the official record shows the consolidated article. The Sharia Procedure Law shows the limit of the remedy. Its consolidated text is accurate and it is misleading, because an instrument that has lost a third of its articles to repeal looks, consolidated, much like an instrument that has lost none. What a reader needs is not only the current text but the fact of its distance from the enacted one, and that is a presentational choice available today at no legislative cost.

Reverse references are the missing tool. If cross-instrument references are the exposed ones, and they are exposed because a drafter cannot see them, then the intervention is to make them visible. A reverse index — for any article, the list of provisions elsewhere that cite it — is computable from a corpus of the kind used here, and consulting it at the point of amendment would convert an invisible risk into a checklist. This is the same instrument the companion citation study proposed for a different purpose, and it is worth noting that one artefact serves both.

‘Amended’ is too coarse a label to publish alone. Section 6 found amendments ranging from a substituted phrase to the replacement of ninety per cent of an article’s wording, all carrying the same label. Where the superseded text exists in the record, showing it — or even showing a similarity indicator — costs nothing and tells a reader whether the amendment they are looking at is worth reading.

For legal information systems. Any system answering questions over statutory text is answering from a snapshot. The results here say where the snapshot decays fastest: not evenly, but in a small number of heavily-cited instruments, and in bursts when a large amending decree lands. A refresh policy that treats all instruments alike is doing too much work on the 58.8 per cent that never change and too little on the handful that carry most of the movement.

9 Limitations

The status layer is a transcription, not an audit. Statuses come from what the official record exposed for each article at the time the corpus was built. Where the record is silent about an

amendment, the article counts as original here. The direction of that error is one-way: churn is understated, not overstated.

Coverage is partial and uneven. 272 of 291 tracks carry statuses, and the 19 that do not are not a random sample. More importantly, the depth of the amendment history varies by source: some instruments expose an event log, some a prior version, most nothing at all. Every measure states the subset it was computed on.

The magnitude sample is small and selected. Eighty-seven articles, 11.9 per cent of amended articles, chosen by whether their source happened to publish a prior version. It supports the claim that amendments vary enormously in magnitude. It does not support any estimate of how the other 88 per cent are distributed.

The reliance gradient is uncontrolled. As section 7 states, age is a live alternative explanation and cannot be tested from this corpus.

Dates are too sparse for a time series. Only 175 of 759 article-amendment pairs carry a Gregorian date — 23 per cent — and the instruments that carry dates are not representative. The apparent concentration of dated amendments in a single year is an artefact of which sources expose dates, and no year-on-year analysis is attempted here.

Reference resolution is automatic. The cross-reference layer is built by pattern extraction, and the companion citation study found that a small share of resolved edges were misresolutions. Section 7's inter-instrument figure rests on 86 references, so a handful of bad resolutions would move it.

10 Conclusion

Saudi legislation changes little in aggregate and very unevenly in distribution. Of 13,089 articles, 973 are no longer in their original form; of 272 instruments, 160 have never changed at all, while ten hold a third of all the change. Instruments do not merely change by different amounts, they change in different ways — hollowed out by repeal, extended by addition, or tuned in place — and the three are different legal events that a single count would merge.

Two of the results bear on how a legal system is used rather than on how it is written. The instruments other instruments rely on are the least stable, though this article cannot separate that from ageing. And references from one instrument into another are twice as exposed to changed text as an instrument's references to itself — an asymmetry with a mechanism a drafting process could address, because it exists only because nobody can see the references pointing at what they are about to amend.

The Law of Sharia Procedure remains the clearest case. It is in force, it is cited, its numbering is intact, and seventy-five of its articles say nothing. Nothing in how it is published conveys that. Now, at least, it can be counted.

Acknowledgements and declarations

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Competing interests. The author declares no competing interests.

Ethics approval. Not applicable. The study involves no human participants, no animal subjects and no personal data; it analyses published national legislation only.

Data availability. The corpus and the analysis and figure scripts that reproduce every number and figure reported here are openly available at <https://github.com/Abdullah-m7/saudi-legal-corpus-ai> and archived on Zenodo under the MIT licence (10.5281/zenodo.22019183; concept DOI 10.5281/zenodo.22019182). The companion studies referred to above — describing the corpus, its citation network, and its definitional lexicon — are under review elsewhere.

Declaration of generative AI use. *Tool:* Claude (Anthropic), version ANONVERSION. *How it was used:* as a research assistant throughout — to write the analysis and figure scripts that produce every number and figure reported here, to draft and revise the manuscript text, and to run adversarial checks against its own claims, which caught several errors before submission. *Reason for use:* the study required building and auditing a reproducible analysis pipeline over a 13,089-article corpus, work that is not feasible by hand at that scale for a single researcher. The author set the research question, directed the analysis, read and verified the outputs against the underlying records, and is responsible for the content. Every number and figure is regenerated by the openly published scripts from the openly published corpus, so that any reader can reproduce them independently of how they were produced.

CRedit roles. A. Almohammed: conceptualisation, data curation, methodology, software, formal analysis, investigation, writing — original draft, writing — review and editing.

Biographical note. Abdullah Almohammed is an independent researcher working on legal informatics and the empirical study of legislation. He built the Saudi Legal Corpus for AI, an openly licensed, article-level, provenance-annotated corpus of Saudi Arabian legislation covering 290 instruments and 15,689 article records, in which every article is recorded against an identified official source. His current work uses that corpus to ask questions about a legal system that are ordinarily argued from examples: how its instruments cite one another, whether they share a vocabulary, and — in this article — what changes in them over time.

Legal status. This article analyses a non-official research corpus. Nothing here constitutes legal advice, and the binding text of any instrument is the Arabic original published in *Umm al-Qura*.